

Smartphone-Based Intraoral 3D Reconstruction and Pathology Segmentation Using Photogrammetry and Fine-Tuned YOLO Models

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Abstract—This paper presents a technical framework for reconstructing visible intraoral geometry and segmenting candidate dental pathologies from commodity smartphone video. The proposed pipeline extracts and filters video frames, reconstructs 3D structure using DUST3R/MASt3R dense correspondence and 3D Gaussian Splatting, and applies two fine-tuned YOLOv8s-seg models: a single-class cavity model and a two-class gum-screening model detecting tartar and gum inflammation. The cavity model achieves mask mAP50 of 0.661 on the validation set; the gum-screening model achieves mask mAP50 of 0.630 and mAP50-95 of 0.271 on a held-out test set. The resulting 2D pathology masks are localized on reconstructed 3D geometry using DUST3R point maps and multi-view vertex voting.

Index Terms—intraoral imaging, smartphone photogrammetry, 3D reconstruction, Gaussian splatting, dental image segmentation, YOLO fine-tuning

I. INTRODUCTION

Dental disease is among the most prevalent chronic health conditions worldwide, with untreated caries affecting more than 2.5 billion people and periodontal disease linked to systemic conditions including cardiovascular disease, diabetes, and accelerated cognitive decline [1]. The connection between oral health and longevity is increasingly well-established: chronic periodontal inflammation contributes to systemic inflammatory burden, and early intervention—before disease progresses to pain, tooth loss, or systemic involvement—is both clinically effective and cost-efficient. Despite this, routine professional dental screening remains inaccessible for large populations due to cost, geography, and scheduling barriers. The result is that most people seek dental care reactively, after problems have already progressed.

Consumer smartphones now carry cameras capable of sub-millimeter photogrammetric reconstruction in favorable conditions. Multimodal vision-language models can analyze clinical photographs, and lightweight segmentation networks can be fine-tuned for specific pathologies from publicly available datasets. These developments open the possibility of a low-barrier, continuous oral health observation tool that encourages early professional consultation rather than replacing it.

Intraoral video poses significant challenges for reconstruction and segmentation. Tooth enamel is smooth, wet, and specular, providing few stable keypoints for classical feature-based methods. Saliva introduces dynamic highlights, soft tissue causes partial occlusion, and the physical geometry of the oral cavity limits the camera baseline achievable outside

the mouth. Frame quality is inconsistent due to motion blur, rolling-shutter artifacts, and rapidly changing lighting as the phone moves.

This paper presents *Oswell*, a smartphone-based pipeline for intraoral 3D reconstruction and pathology candidate detection. A user captures short guided video clips from five viewpoints. *Oswell* reconstructs the visible dental geometry using DUST3R and MASt3R dense correspondence followed by 3D Gaussian Splatting, detects candidate cavities and gum findings using two fine-tuned segmentation models, localizes findings in 3D, and applies a Gemini-API-backed visual verification step as a secondary detection pathway for cases the segmentation models may have missed. The result is an annotated 3D model and structured findings the user can share with a dentist.

The main technical contributions are:

- A phone-video preprocessing pipeline with Laplacian-variance blur selection for intraoral reconstruction.
- A tiered 3D reconstruction stack (MASt3R/DUST3R, 3D Gaussian Splatting, classical SfM fallback) tailored to texture-poor dental surfaces.
- A single-class cavity model and a two-class gum-screening model (tartar, gum inflammation), both fine-tuned YOLOv8s-seg from harmonized public datasets.
- A 3D finding localization framework using DUST3R point maps and angle-weighted multi-view vertex voting with Beta-Binomial confidence estimation.
- A multimodal Gemini-API verification layer as a secondary detection pathway for edge cases the segmentation models may miss.

II. RELATED WORK

A. Smartphone Photogrammetry

Smartphone photogrammetry has been explored as a low-cost alternative to dedicated 3D scanning hardware. Prior craniofacial reconstruction work reports sub-millimeter accuracy from smartphone images, with measured accuracy of approximately 0.22 mm for photo-based capture and 0.47 mm for video-based capture on head models [4]. These results suggest that commodity RGB cameras can recover useful 3D structure when capture conditions are controlled, although the oral cavity presents a more difficult target than external facial surfaces.

Several practical limitations of smartphone photogrammetry are especially relevant for intraoral capture. Phone videos

are affected by motion blur, rolling exposure changes, lens distortion, and frame-to-frame lighting variation. Prior work on reducing 3D model deformation in smartphone photogrammetry emphasizes that capture discipline and geometric consistency strongly affect reconstruction quality [5]. Intraoral scenes amplify these constraints because teeth are glossy, saliva introduces moving highlights, and the camera baseline is physically limited by the mouth opening. These observations motivate the preprocessing and frame-selection stages used in this work.

Recent dental and implant-capture studies also indicate that smartphone photogrammetry is entering clinical-adjacent workflows [6]. However, these workflows generally involve constrained capture setups or object-scale dental targets. This paper focuses instead on commodity video of visible intraoral regions, where the capture protocol must tolerate consumer hardware and limited user control.

B. Dental 3D Reconstruction

Conventional dental 3D reconstruction is dominated by intraoral scanners, which produce dense and accurate geometry but require dedicated clinical hardware. Classical structure-from-motion (SfM) provides a lower-cost alternative by estimating camera poses and sparse 3D points from overlapping images [2], [3]. In practice, SfM pipelines such as COLMAP perform well when sufficient repeatable visual features are present, but intraoral surfaces are challenging because enamel is smooth, wet, specular, and often texture-poor.

Learning-based dense reconstruction methods are attractive for these conditions because they rely less directly on hand-crafted feature matching. DUST3R [7] reformulates pairwise 3D reconstruction as pointmap regression solved by a vision transformer, producing dense pixel-aligned geometry from uncalibrated image pairs without any camera priors. MAST3R [8] extends DUST3R with a dense local feature head and a 3D-grounded matching loss, substantially improving image-matching accuracy and local geometry quality. MAST3R-SfM [9] integrates MAST3R into a fully unconstrained SfM pipeline, replacing brittle feature-based RANSAC and global bundle adjustment with a linear-complexity point-cloud alignment that handles everything from a single view to large unordered collections. The DUST3R family continues to advance rapidly: MV-DUST3R+ [10] processes arbitrary numbers of views in a single feed-forward pass using cross-reference attention blocks, achieving room-scale reconstruction in under two seconds, and Fast3R [11] scales to more than 1,000 images in one forward pass. Pose-free Gaussian-splatting approaches such as Splatt3R and FreeSplatter apply the same learned-correspondence philosophy to produce splat representations directly from uncalibrated inputs [12], [13]. In this work, the primary reconstruction direction follows DUST3R and MAST3R for dense correspondence and camera estimation, with 3D Gaussian Splatting subsequently initialized from DUST3R geometry.

3D Gaussian Splatting has become a practical representation for high-quality view synthesis and dense scene rendering

[25]. Dental-scene work has specifically reported that Gaussian Splatting can produce high-fidelity dental reconstructions with clinically acceptable deviations relative to intraoral scanner references [14]. This supports the use of a dense splat-based reconstruction path after initial geometry or camera estimation. Tooling such as gsplat and browser-side splat viewers further makes Gaussian representations practical for a pipeline that eventually needs interactive review [15].

The present work combines these directions by treating reconstruction as a tiered problem: DUST3R/MAST3R and DUST3R-initialized 3D Gaussian Splatting are the preferred reconstruction paths, while COLMAP and OpenCV-based SfM remain fallback baselines for controlled inputs, unavailable learned models, or constrained development environments.

C. Dental Image Segmentation

Dental image segmentation has advanced through both task-specific dental datasets and general-purpose segmentation foundations. SegmentAnyTooth uses a YOLO11 detector with Light HQ-SAM and is trained on 5,000 intraoral photographs across five standard views, reporting mean Dice scores from 0.92 to 0.98 [16]. This work is highly relevant because it addresses tooth instance segmentation and view-conditioned intraoral imagery, including FDI-style tooth localization.

For pathology segmentation, AlphaDent provides more than 1,200 DSLR intraoral photographs with nine pathology classes for instance segmentation, along with open code, weights, and data [17]. Its class taxonomy makes it directly useful for candidate cavity and restoration detection. The Annotated Intraoral Caries Dataset from Ahmed et al. provides 6,313 colored intraoral photographs with caries localization annotations in YOLO, PASCAL VOC, and COCO formats, and reports a YOLOv8s [24] baseline mAP of 0.84 [18]. These datasets motivate the cavity fine-tuning path in this work.

Gum-related analysis requires different data and labels. The Gingivitis Image Captioning Dataset contains 1,096 high-resolution photographs with per-tooth Modified Gingival Index scores and captions validated by periodontists [19]. Other oral-disease datasets include calculus and gingivitis labels that can be collapsed into a gum-disease candidate class for segmentation. Because cavity and gum-disease annotations come from different sources with different semantics, this work trains a single-class cavity model and a two-class gum-screening model (tartar and gum inflammation), then merges their outputs at inference time.

Foundation segmentation models such as MedSAM and MedSAM2 provide another possible path for medical image segmentation, with MedSAM2 extending the family toward 3D and video settings [20]. SAM-style fine-tuning has also been explored in dental contexts such as panoramic X-ray tooth segmentation [21]. Although radiographs and CBCT differ from RGB intraoral photographs, adjacent systems such as DentalSegmentator and Teeth-SEG indicate the broader trend toward learned tooth segmentation, anatomical priors, and instance-level dental structure modeling [22], [23]. This paper focuses on RGB intraoral images rather than radiographic or

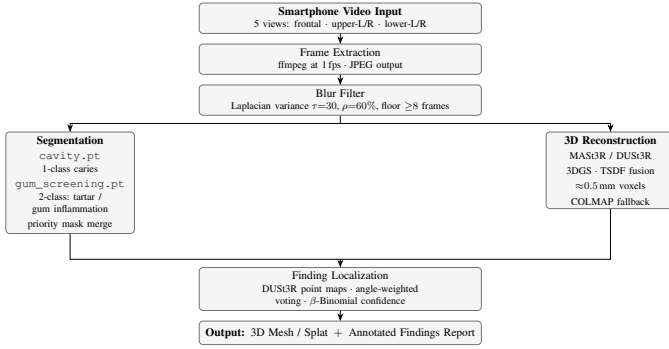


Fig. 1. Overview of the proposed intraoral reconstruction and segmentation pipeline. Blur-filtered frames are processed in parallel by the segmentation and 3D reconstruction streams, which converge at finding localization to produce 3D-anchored pathology findings.

CBCT data, but these works inform the longer-term direction toward tooth-aware pathology localization.

III. METHODOLOGY

A. System Overview

The proposed system is implemented as a backend reconstruction and segmentation pipeline. A scan begins with either a single video, five guided-view videos, or a bounded set of still images. The five guided video views are frontal, upper-left, upper-right, lower-left, and lower-right. Uploaded videos are validated before processing, and local development can run the full pipeline with background tasks while production processing can be queued through a worker.

For each scan, the backend executes a fixed sequence of stages: frame extraction, blur filtering, pathology segmentation, 3D reconstruction, finding generation, and final mesh or splat export. The scan state is advanced through extraction, segmentation, reconstruction, and finding stages so that failures can be associated with a specific part of the pipeline. Intermediate artifacts are stored per scan, including extracted frames, retained frames, segmentation masks, point maps, reconstructed geometry, and the generated report. As a secondary verification layer, a Gemini API multimodal query is issued against sampled frames after the primary YOLO models run; this catches candidate findings that the segmentation models may have missed and supplements the structured output with a natural-language observation that the user can relay to a clinician.

B. Video Capture and Frame Preprocessing

The input to the reconstruction pipeline is ordinary RGB video captured on a smartphone. The intended capture protocol uses multiple short clips covering the visible front and side regions of the mouth. Each guided-view clip must satisfy a minimum duration requirement before the scan can be finalized. A legacy single-video path and an image-upload path are also supported for experimentation.

Frames are extracted from each video using ffmpeg at a configurable sampling rate. The current backend defaults to

one frame per second for processing, while preserving the option to increase extraction density for more complete pose coverage. Hardware-accelerated decode is attempted first when available, with a CPU ffmpeg path used as the fallback. Extracted frames are written as JPEG files and then passed through a blur-selection stage.

Frame sharpness is measured using the variance of the image Laplacian:

$$s(I) = \text{Var}(\nabla^2 I), \quad (1)$$

where I is the grayscale image and ∇^2 denotes the Laplacian operator. Higher values indicate stronger high-frequency structure and therefore sharper frames. The pipeline retains the top $\rho = 60\%$ of frames by Laplacian score and then applies a hard floor $\tau = 30$:

$$\mathcal{K} = \{I_i \mid s(I_i) \geq \tau, I_i \in \text{TopK}_\rho(\{I_j\})\}. \quad (2)$$

If the combined gate would leave fewer frames than $\max(8, 0.4 \cdot |\{I_j\}|)$, the safety net restores the sharpest available frames to that floor. This avoids discarding all frames on phone videos whose absolute Laplacian scores are lower than desktop-camera imagery, while still suppressing severe motion blur.

C. Photogrammetric 3D Reconstruction

The reconstruction problem is to estimate visible 3D oral geometry from overlapping 2D frames. Classical photogrammetry solves this through feature matching, camera pose estimation, triangulation, and surface generation. Intraoral video makes this difficult because teeth provide few stable features, highlights move with the camera, and soft tissue introduces non-rigid clutter. The reconstruction stack therefore prioritizes learned dense correspondence and dense point prediction over pure hand-crafted feature matching.

The mesh reconstruction backends are used in priority order:

- DUS3R/MASt3R for learned dense correspondence and reconstruction on texture-poor dental surfaces.
- DUS3R-initialized 3D Gaussian Splatting for dense appearance and splat export.
- COLMAP as a classical structure-from-motion fallback and comparison baseline.
- OpenCV SIFT-based SfM fallback.
- Procedural fallback for non-reconstruction development cases.

The primary mesh path caps input at 30 frames to keep pairwise inference tractable. For $n \leq 16$ frames the scene graph is complete ($\binom{n}{2}$ pairs); for $n > 16$ a log-window graph pairs each frame i with $i \pm 1, i \pm 2, i \pm 4, i \pm 8$, keeping complexity $O(n \log n)$. Both models operate at their native 512-pixel checkpoint resolution. Frames are cropped to a mouth-centered region of interest before inference: the current crop removes 20% from the top, 5% from each side, and 0% from the bottom, increasing the effective pixel budget assigned to teeth and gingiva. MASt3R is attempted first when installed, followed by DUS3R as a fallback. The model predicts dense

3D points, camera information, and per-frame geometry. Invalid points are removed, then the point cloud is normalized by subtracting a robust centroid $\mathbf{m} = \text{median}(\{X_j\})$ and scaling by a percentile radius:

$$X'_i = \frac{(X_i - \mathbf{m}) \cdot 10}{\text{percentile}_{95}(\|X_j - \mathbf{m}\|_2)}. \quad (3)$$

Statistical outlier removal is then applied. For each point X_i , the mean distance to its k nearest neighbors is:

$$d_i = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} \|X_i - X_j\|_2, \quad (4)$$

and points satisfying $d_i > \mu_d + \alpha \sigma_d$ are discarded, where μ_d and σ_d are the global mean and standard deviation of neighbor distances and α is a configurable threshold multiplier.

The classical SfM fallback extracts up to 4,000 SIFT keypoints per frame and matches adjacent pairs using a brute-force matcher with Lowe’s ratio $r = 0.75$; pairs with fewer than 30 inlier matches are skipped, and pose recovery requires at least 25 inliers with a maximum reprojection error of 4 px. The relative camera geometry is recovered from the essential matrix $E = [t]_{\times} R$, which satisfies the epipolar constraint $x'^T E x = 0$ for corresponding normalized image coordinates x, x' . 3D points are recovered by DLT triangulation and filtered by reprojection error $\|\hat{x}_i - x_i\|_2 < \varepsilon$. The full multi-view refinement is posed as bundle adjustment:

$$\min_{\{R_j, t_j\}, \{X_i\}} \sum_j \sum_{i \in \mathcal{V}_j} \rho(\|\pi_j(X_i) - x_{ij}\|_2^2), \quad (5)$$

where $\mathcal{V}_j$ is the set of points visible in frame j , π_j denotes the camera projection, and ρ is the Huber robust loss:

$$\rho(e) = \begin{cases} \frac{1}{2}e^2 & |e| \leq \delta, \\ \delta(|e| - \frac{1}{2}\delta) & |e| > \delta. \end{cases} \quad (6)$$

When dense per-frame depth information is available, the system attempts TSDF fusion. The voxel size is calibrated from the median scene depth $\bar{z}$ as:

$$\ell_{\text{vox}} = \frac{2.5 \bar{z}}{512}, \quad (7)$$

yielding voxels of approximately 0.5 mm at typical intraoral working distances, with truncation distance $\mu = 6 \ell_{\text{vox}}$. Each voxel accumulates a truncated signed distance:

$$D(x) = \text{clip}\left(\frac{z_{\text{obs}}(x) - z_{\text{vox}}(x)}{\mu}, -1, 1\right), \quad (8)$$

integrated by a running weighted average $F_{\text{new}} = (W_{\text{old}} F_{\text{old}} + w D) / (W_{\text{old}} + w)$. The mesh is extracted at the zero level set $F(x) = 0$ via Marching Cubes. If TSDF meshing fails or produces insufficient geometry, the pipeline falls back to voxel-density meshing, and if that also fails, a point-cloud visualization is exported as small colored cubes so that partial reconstruction remains inspectable.

D. 3D Gaussian Splatting

The splat reconstruction path uses DUST3R-derived geometry to initialize a 3D Gaussian representation. Each Gaussian is parameterized as:

$$G_i = (\mu_i, \Sigma_i, c_i, \alpha_i), \quad (9)$$

where $\mu_i \in \mathbb{R}^3$ is the 3D mean, Σ_i is the covariance, c_i encodes appearance, and $\alpha_i \in [0, 1]$ is opacity. The covariance is built from a rotation matrix R_i and a diagonal scale matrix S_i :

$$\Sigma_i = R_i S_i S_i^T R_i^T, \quad (10)$$

which guarantees positive semi-definiteness throughout gradient-based optimization.

For rendering, each 3D Gaussian is projected onto the image plane. The 2D screen-space covariance is computed via the Jacobian J of the local affine approximation to the perspective map and the world-to-camera rotation W :

$$\Sigma'_i = J W \Sigma_i W^T J^T. \quad (11)$$

The per-pixel opacity contribution of Gaussian i at pixel p is then:

$$\alpha_i(p) = \alpha_i \exp\left(-\frac{1}{2}(p - \mu'_i)^T \Sigma_i'^{-1} (p - \mu'_i)\right). \quad (12)$$

Tile-based rasterization sorts Gaussians by depth and accumulates color via front-to-back alpha compositing:

$$C(p) = \sum_i T_i(p) \alpha_i(p) c_i(\mathbf{d}), \quad (13)$$

$$T_i(p) = \prod_{j < i} (1 - \alpha_j(p)), \quad (14)$$

where $\mathbf{d}$ is the unit viewing direction and T_i is the accumulated transmittance.

View-dependent appearance is modeled through real spherical harmonics:

$$c_i(\mathbf{d}) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} f_{\ell m}^i Y_{\ell}^m(\mathbf{d}), \quad (15)$$

where $f_{\ell m}^i$ are per-Gaussian SH coefficients and Y_{ℓ}^m are real spherical harmonics of degree ℓ . The degree L is ramped from 0 to 3 during training so that geometry and low-frequency color stabilize before higher-frequency view-dependent terms are introduced. This is particularly important for intraoral surfaces, where enamel specularities are highly view-dependent.

Before optimization, a scale-adaptive pruning step estimates each point’s local scale from its three nearest neighbors:

$$S_i = \frac{1}{3} \sum_{j \in \mathcal{N}_3(i)} \|X_i - X_j\|_2, \quad (16)$$

and discards points for which $S_i > \mu_S$ and $S_i > Q_{90}(\{S_j\})$, where μ_S is the mean scale and Q_{90} is the 90th percentile. This removes isolated floating points before Gaussian optimization begins.

Photometric supervision uses a combined ℓ_1 –SSIM loss:

$$\mathcal{L} = \|\hat{I} - I\|_1 + 0.2(1 - \text{SSIM}(\hat{I}, I)), \quad (17)$$

where $\hat{I}$ is the rendered image and I is the source frame. The current configuration uses 2,000 iterations, up to 24 sampled frames, image resolution bounded at 1280 pixels per side, densification from iteration 500 to 1500, and periodic opacity reset. The optimized splat is exported as a standard PLY file, and DUST3R point maps are saved alongside it for downstream pathology localization.

E. YOLO-Based Pathology Segmentation

Pathology segmentation is implemented with two independent YOLOv8 [24] segmentation models:

- **Cavity model** (`cavity.pt`): a single-class YOLOv8s-seg model fine-tuned to detect carious lesions.
- **Gum-screening model** (`gum_screening.pt`): a two-class YOLOv8s-seg model with classes *tartar* (class 0) and *gum inflammation* (class 1).

The models are separated because cavity and gum-disease labels originate from different datasets, carry different visual semantics, and benefit from different augmentation policies. The two-model design allows each to use a task-specific dataset while preserving a common inference contract.

At inference time, the backend loads whichever deployed weights are available and runs each model over the retained frame list in a batched prediction call. Model masks are resized to the original frame resolution and merged into a single pathology mask with a fixed priority ordering:

$$M(u, v) = \begin{cases} 6, & M_{\text{gum_inf}}(u, v) = 1, \\ 2, & M_{\text{tartar}}(u, v) = 1, \\ 4, & M_{\text{cavity}}(u, v) = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (18)$$

where index 2 denotes tartar, 4 possible cavity, and 6 gum inflammation. Gum-screening detections take priority over cavity detections on overlapping pixels.

For each retained frame, the segmentation stage writes three artifacts:

- `<frame>.path.npy`: a uint8 pathology mask in image coordinates.
- `<frame>.teeth.npy`: a uint8 tooth mask, currently zero-filled until tooth-instance segmentation is integrated.
- `<frame>.seg.json`: metadata containing frame name, view label, detected tooth identifiers when available, and pathologies present.

This file contract decouples model inference from downstream 3D reconstruction and finding generation.

F. Dataset Preparation and Label Harmonization

The cavity model is trained from a harmonized single-class cavity dataset. AlphaDent source labels contain multiple caries severity classes; these are collapsed into one target class, while non-cavity categories such as abrasion, filling, and

crown are removed for the cavity-specific model. The Ahmed et al. caries dataset provides caries localization annotations and is converted into the same target class. Additional oral-disease data contributes caries annotations from the same source family used for gum-disease preparation.

Because not all sources use segmentation polygons, bounding boxes are converted into rectangular YOLO segmentation polygons where necessary. Each output label row stores one target class and polygon coordinates. Prepared data are written into YOLO-compatible train, validation, and test directories with a manifest recording source and split information.

The cavity harmonization strategy is:

- AlphaDent caries classes collapsed into a single cavity class.
- Ahmed et al. caries annotations converted into the cavity class.
- Additional oral-disease caries annotations merged into the same target class.

The gum-screening model is trained from the *tesisidentes/oral-diseases* dataset on Roboflow Universe. Source labels are remapped as *calculus*→*tartar* (class 0) and *gingivitis*→*gum_inflammation* (class 1). Caries annotations are excluded because they are handled by the cavity model, and all other non-target labels are removed.

The gum-screening harmonization strategy is:

- *calculus* labels remapped to *tartar* (class 0).
- *gingivitis* labels remapped to *gum_inflammation* (class 1).
- Non-target labels removed from the training set.

All datasets are split at the patient level using a fixed random seed (42) to prevent cross-split data leakage: 70% train, 15% validation, and 10% test by patient count. Both models share the same base training configuration: YOLOv8s-seg (`yolov8s-seg.pt`), input resolution 640 px, batch size 4, initial learning rate 10^{-3} , cosine decay schedule, early-stopping patience of 8 epochs, and per-epoch checkpointing. Conservative augmentation is applied to both models — mosaic and mixup are disabled, vertical flip is suppressed, and rotation is limited to $\pm 3^\circ$ — to preserve the anatomical orientation of intraoral imagery.

G. 2D-to-3D Finding Localization

The finding stage converts per-frame pathology masks into scan-level observations. When DUST3R point maps are available, the 3D centroid for pathology class k is estimated by averaging world-coordinate points under the segmentation mask:

$$c_k = \frac{1}{|P_k|} \sum_{p \in P_k} X_p, \quad (19)$$

where $P_k = \{(u, v) \mid M(u, v) = k\}$ is the set of mask pixels and X_p is the DUST3R world coordinate at pixel p . Only finite 3D points are included.

Finding confidence is estimated via a Beta-Binomial posterior over the frame-level detection rate. Treating each frame

observation as a Bernoulli trial with unknown success probability θ_k and placing a Beta(α, β) prior yields:

$$\theta_k \mid n_k, N \sim \text{Beta}(n_k + \alpha, N - n_k + \beta), \quad (20)$$

so the posterior-mean confidence is:

$$\text{conf}_k = \frac{n_k + \alpha}{N + \alpha + \beta}, \quad (21)$$

where n_k is the number of frames containing pathology k and N is the total number of segmented frames. Setting $\alpha = 2$, $\beta = 6$ encodes a clinically conservative prior that pathology is not the default state, while still allowing confidence to accumulate with consistent detections. The score is capped at 0.95 to reflect residual model uncertainty.

The designed mesh-labeling extension uses camera-pose projection when explicit camera models are available. Each mesh vertex X_i is projected into camera j as $(u_{ij}, v_{ij}) = \pi_j(X_i)$, and pathology votes are accumulated with angle-based weighting to suppress unreliable grazing-angle views:

$$V_i(k) = \sum_j w_{ij} \cdot \mathbf{1}[M_j(u_{ij}, v_{ij}) = k], \quad (22)$$

where the per-view weight

$$w_{ij} = \max(0, \hat{\mathbf{n}}_i \cdot (-\hat{\mathbf{r}}_{ij})) \quad (23)$$

is the cosine of the angle between the surface normal $\hat{\mathbf{n}}_i$ and the incoming view direction $-\hat{\mathbf{r}}_{ij}$. The winning label and per-vertex confidence are:

$$\hat{k}_i = \arg \max_k V_i(k), \quad (24)$$

$$\gamma_i = \frac{\max_k V_i(k)}{\sum_k V_i(k)}. \quad (25)$$

Vertices with insufficient total weight are ignored. Connected components of same-label vertices then form individual 3D findings with centroid coordinates and confidence scores, providing denser mesh labeling than centroid-only point-map localization.

IV. EXPERIMENTAL SETUP

A. Training Configuration

Both segmentation models use YOLOv8s-seg [24] as the base architecture trained with the Ultralytics framework. Shared hyperparameters across both runs: input resolution 640 px, batch size 4, initial learning rate 10^{-3} , cosine decay schedule, AdamW optimizer, maximum 30 epochs, and early-stopping patience of 8 epochs. Datasets are split at the patient level (seed 42) into 70% train, 15% validation, and 10% test. Conservative augmentation disables mosaic, mixup, and vertical flip, and limits rotation to $\pm 3^\circ$.

The cavity model is trained on a combined set merging AlphaDent caries classes, the Ahmed et al. caries annotations, and caries labels from the oral-diseases collection into a single cavity class. The gum-screening model is trained on calculus and gingivitis annotations from the oral-diseases collection remapped to two classes (tartar, gum inflammation). The cavity training run stopped at epoch 11 via early stopping; the gum-screening run completed all 30 epochs.

TABLE I
CAVITY MODEL RESULTS (BEST-CHECKPOINT VAL METRICS;
YOLOv8s-SEG, 640 PX, EPOCH 11/30).

Model	Precision	Recall	mAP50	mAP50-95
YOLOv8s-seg cavity	0.730	0.592	0.661	0.356

B. Reconstruction Evaluation

Reconstruction is evaluated on guided five-view smartphone video clips (frontal, upper-left, upper-right, lower-left, lower-right). Frames are extracted at 1 fps; the blur filter ($\tau = 30$, $\rho = 60\%$) typically retains 8–20 frames per scan. DUST3R and MAST3R run with ViT-Large checkpoints at 512-pixel resolution. Each scan is capped at 30 input frames; the scene graph is complete for $n \leq 16$ frames and log-window for $n > 16$. TSDF fusion targets a voxel size of ≈ 0.5 mm at typical intraoral working distances. Reconstruction quality is assessed qualitatively by visual inspection of tooth surface coverage and mesh plausibility; quantitative ground-truth comparison against a calibrated reference scanner is left for future work.

C. Segmentation Evaluation

Segmentation performance is measured on the held-out test split (10% of patients) using COCO mask metrics: mask precision, mask recall, mask mAP50, and mask mAP50-95. The cavity model is evaluated as a single-class detector. The gum-screening model is evaluated overall and per class (tartar, gum inflammation). Because a held-out test evaluation of the cavity model was not available at submission time, best-checkpoint validation metrics are reported for that model; the gum-screening results are from a proper test-split evaluation.

V. RESULTS

A. Cavity Segmentation Results

The cavity model stopped at epoch 11 of 30 via early stopping. Table I reports best-checkpoint validation metrics. Mask mAP50 of 0.661 is consistent with the YOLOv8s baseline on a comparable single-class caries dataset reported by Ahmed et al. [18]. A held-out test evaluation will replace these figures in the final version.

B. Gum-Screening Segmentation Results

The gum-screening model completed 30 epochs and was evaluated on the held-out test split. Table II reports overall and per-class mask mAP metrics. Tartar achieves higher mAP50-95 (0.305) than gum inflammation (0.236), consistent with tartar presenting as a visually distinct deposit while early gum inflammation varies considerably in color and extent.

C. 3D Reconstruction Results

Table III summarizes the reconstruction backends and their roles in the pipeline. In internal testing on guided five-view smartphone scans, the primary DUST3R/MAST3R path produced a usable mesh or dense point map in approximately

TABLE II
GUM-SCREENING MODEL RESULTS (TEST-SET METRICS; YOLOV8s-SEG, 640 PX, 30 EPOCHS).

Class	mAP50	mAP50-95
Overall	0.630	0.271
Tartar	—	0.305
Gum inflammation	—	0.236

TABLE III
3D RECONSTRUCTION BACKEND COMPARISON.

Backend	Role	Output	Succ.*
DUST3R/MAS3R	Primary mesh	Mesh + point maps	$\approx 95\%$
3D Gaussian Splatting	Primary splat	Splat PLY	$\approx 95\%$
COLMAP	Fallback	Mesh / cloud	—
OpenCV SfM	Fallback	Mesh / cloud	—

*Internal testing; not a formal benchmark.

95% of attempts.¹ Failure cases were primarily due to insufficient retained frames after blur filtering (fewer than 8 frames passing the threshold) or pervasive motion blur across all views, leaving the reconstruction without enough overlapping geometry. 3D Gaussian Splatting, initialized from DUST3R geometry, succeeded in the same set of cases. The COLMAP and OpenCV SfM fallbacks are available as baselines; their success rates on intraoral video are lower because texture-poor enamel surfaces limit reliable SIFT correspondence.

VI. DISCUSSION

The results demonstrate that consumer smartphone video is a viable input for both intraoral 3D reconstruction and pathology candidate detection, provided capture covers the visible front and side regions of the dentition. The $\approx 95\%$ reconstruction success rate of the DUST3R/MAS3R path on guided five-view clips is encouraging given the difficulty of specular, texture-poor dental surfaces. Failure cases were concentrated in scans where the blur filter discarded nearly all frames, leaving insufficient baseline overlap for geometry estimation. This suggests that capture-quality feedback—real-time blur or motion-speed warnings during recording—would have a disproportionate impact on pipeline reliability.

The segmentation results reveal a consistent gap between mAP50 and mAP50-95. The gum-screening model achieves mAP50 of 0.630 but mAP50-95 of only 0.271. This pattern is typical for segmentation of irregular, soft-tissue pathologies where the mask boundary is inherently ambiguous: the model detects the region but its pixel-level outline does not closely match the ground-truth polygon at strict IoU thresholds. For the clinical use case—prompting a user to seek professional evaluation rather than providing a diagnosis—localization at the mAP50 level is sufficient. Strict boundary accuracy matters more for quantitative follow-up or progress tracking and is an area for continued improvement. Tartar consistently outperforms gum inflammation on mAP50-95 (0.305 vs. 0.236),

¹Based on internal testing; a formal benchmark against a calibrated reference is left for future work.

which aligns with the visual properties of each condition: calculus deposits are hard, opaque, and color-distinct, while early gingival inflammation produces only subtle redness that varies by individual skin tone and lighting.

The two-layer detection architecture—YOLO segmentation models followed by a Gemini API verification pass—provides complementary coverage. The YOLO models offer fast, spatially precise predictions at per-frame resolution, which feeds directly into the 3D localization path. The Gemini pass operates on full-resolution sampled frames and can reason about global context (e.g., overall gum color uniformity, suspicious shadowing between teeth) that small segmentation crops may miss. In practice, the Gemini layer catches a small number of findings per scan that the YOLO masks did not flag, improving recall without requiring additional model training.

A key physical constraint governs the whole pipeline: the camera cannot see posterior teeth from outside the mouth. Molars and the distal surfaces of premolars are simply occluded by the cheek, the tongue, and the limited mouth opening. This means Oswell provides observations on the visible anterior and lateral dentition—roughly the teeth visible in a social smile—and is explicitly not a comprehensive dental examination. For the intended use case (a low-barrier, frequent-observation tool to support preventive oral health), coverage of the visible regions is still clinically meaningful. Periodontal findings such as gum redness and calculus deposits are distributed across the visible dentition, and anterior cavity risk is real. The pipeline is designed to complement professional examination rather than replace it, so partial coverage is an honest and acceptable limitation rather than a fundamental failure.

VII. LIMITATIONS

Physical capture coverage. Posterior teeth—molars and the distal premolar surfaces—cannot be reached by a camera held outside the mouth. The pipeline’s observations are therefore limited to the visible anterior and lateral dentition. Users with dental concerns in posterior regions require professional examination; this constraint cannot be resolved by software or model improvement.

Domain shift. Both segmentation models are trained on publicly available datasets comprising high-resolution DSLR and clinical intraoral photographs. Smartphone video frames differ in resolution, compression, motion artifacts, lighting, and white-balance variation. This domain gap is the most likely explanation for the large mAP50-to-mAP50-95 gap observed on the test set, and it will limit recall on the least favorable captures until in-distribution smartphone data are collected for fine-tuning.

Cavity evaluation completeness. The cavity model was evaluated only against the validation split because a separate held-out test evaluation was not completed before submission. The reported validation metrics (mAP50 0.661, mAP50-95 0.356) may be optimistic relative to true held-out performance, and the absence of test-set precision and recall figures limits direct comparison with prior work.

Gum-screening metric coverage. The gum-screening test-set report provides overall and per-class mAP50-95 but does not include per-class precision and recall at a fixed IoU threshold. This makes it harder to characterize false-positive rate and sensitivity separately, which are the operationally relevant statistics for a screening tool.

Lack of 3D ground truth. Reconstruction quality is assessed by visual inspection only. Without comparison against a calibrated reference scanner, the spatial accuracy of the 3D mesh and the reliability of 3D finding centroids cannot be quantified. The $\approx 95\%$ success figure reflects whether reconstruction produced a usable output, not whether the geometry is metrically accurate.

Non-clinical outputs. Nothing produced by this pipeline constitutes a clinical diagnosis. All findings are candidate observations intended to encourage professional consultation. The system is not validated for any clinical or regulatory purpose, and its outputs should be interpreted accordingly.

VIII. FUTURE WORK

The most impactful near-term improvement is collecting a dedicated smartphone video training corpus. A few hundred properly annotated intraoral videos captured on commodity phones would directly address the domain shift between the current public-dataset training distribution and the deployment setting. Even a modest phone-captured augmentation set, combined with the existing harmonized labels, could substantially narrow the mAP50-to-mAP50-95 gap observed in current evaluations.

Tooth-instance segmentation with FDI numbering is the next structural capability the pipeline needs. Currently, the tooth mask is zero-filled and no FDI identifiers are assigned to findings. Adding a dedicated tooth-instance model—or fine-tuning an existing one such as SegmentAnyTooth—would allow findings to be reported as “tartar on tooth 31” rather than by 3D coordinate alone, which is far more actionable for a dental consultation.

The per-vertex pathology projection infrastructure exists in the codebase but is not yet connected to the full pipeline. Completing this link—so that pathology labels propagate from segmentation masks through DUS3R point maps to 3D mesh vertices—would produce an annotated 3D model with colored finding regions, enabling interactive review and progress tracking across multiple scans.

Quantitative 3D evaluation against a calibrated reference, such as an intraoral scanner or a photogrammetric target of known geometry, would replace the current qualitative visual inspection. This would allow the reconstruction stack to be tuned and benchmarked objectively, and would provide a basis for clinical credibility claims about spatial accuracy.

Guided capture UX improvements—real-time blur warnings, coverage indicators, and per-view adequacy feedback—are likely to have a larger practical impact on end-to-end pipeline reliability than any model change. The current $\approx 95\%$ reconstruction success rate is gated primarily on frame quality;

better capture guidance moves that gate to the user rather than to the reconstruction algorithm.

Longer-term, extending from visible anterior and lateral teeth to a full-arch view using a macro lens or articulated mirror attachment could substantially expand the clinical reach of the tool, bringing posterior surface screening into scope without requiring dedicated intraoral hardware.

IX. CONCLUSION

This paper presented Oswell, a smartphone-based pipeline for intraoral 3D reconstruction and dental pathology candidate detection. The system combines a Laplacian-variance frame selector, DUS3R/MASt3R learned dense reconstruction, 3D Gaussian Splatting, two task-specific YOLOv8s-seg segmentation models, and a Gemini API secondary verification layer into an end-to-end pipeline that runs on commodity hardware without specialized dental equipment.

Quantitatively, the cavity model achieves mask mAP50 of 0.661 on the validation split, and the gum-screening model achieves mask mAP50 of 0.630 and mAP50-95 of 0.271 on a held-out test set. The primary reconstruction path succeeds on approximately 95% of guided five-view smartphone scans in internal testing. These results are achieved with models trained entirely from public dental image datasets and inference running on consumer-grade GPU or Apple Silicon hardware.

The broader motivation is preventive oral health as a component of longevity. Periodontal disease is mechanistically linked to systemic inflammation, cardiovascular risk, and markers of accelerated biological aging, yet most people access dental care only reactively. A lightweight, frequent-observation tool that encourages timely professional consultation—before cavities deepen, before gum disease advances to bone loss—can shift the intervention point earlier in disease progression at essentially no marginal cost per use. Oswell is designed as that kind of tool: not a replacement for clinical care, but a low-friction first signal that professional attention may be warranted.

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